

MIE 402 Lab 1 Report Instructions

Dynamic Data Sampling and Frequency Analysis | Fall 2026

This report explains how sampling rate changes the recorded representation of sound. Use your own Moku:Go data from three sound sources and the five required sample-rate cases. Your main conclusion must state which recordings represent the source reliably, which do not, and what evidence supports that judgment.

Required experiment and data set

- Record at least three sounds: a tuning fork, a whistle, and a finger snap, or GTA-approved alternatives.
- For each sound, collect records at 100f, 10f, 3f, 1.4f, and 1f for at least one second. Define f for each source. For the tuning fork, use its labeled fundamental frequency.
- Keep microphone position, input channel, gain, acquisition mode, and source conditions as consistent as practical while changing sample rate.
- Export every recording as a MAT file. Record the requested rate, achieved rate from the time column, duration, source, trial, and relevant settings.
- Before leaving, show the GTA one valid high-rate tuning-fork waveform and a spectrum with a clear peak near the labeled fork frequency. Back up all raw files.

Files to submit

- One group report in PDF, following the current MIE 402 report template and the Canvas assignment.
- The MATLAB analysis file used for your figures, if Canvas requests it.
- Any additional files requested on Canvas. Keep the original MAT files available; upload them only if the assignment requests them.

Canvas controls the due date, file limits, naming rule, and any requirement that differs from this guide.

Questions the report must answer

1. How does lowering sample rate change the time-domain waveform for each source?
2. How does lowering sample rate change the frequency spectrum and the represented peak frequencies?
3. Which components violate the Nyquist condition, and where should their aliases appear?
4. How do mean and RMS change across rates? Why can a plausible RMS value still accompany an incorrect spectrum?
5. How does the replayed sound change? Separate acquisition aliasing from a playback-rate mistake.
6. How do a nearly sinusoidal tuning fork, a harmonic-rich whistle, and a transient snap differ?
7. Which sample-rate choice would you recommend for each source, and what practical margin would you use?

Recommended report organization

Section	What it must do for Lab 1
Abstract	Purpose, three-sound method, most important numerical finding, and conclusion. Maximum 10 lines.
Introduction	Define sampling rate, Nyquist frequency, aliasing, mean, RMS, FFT, and frequency-bin spacing. State the objective and predictions.

Section	What it must do for Lab 1
Method	Identify Moku:Go, microphone, sound sources, connection, rate matrix, duration, settings held constant, MAT export, and analysis steps.
Results	Present selected waveforms, spectra, tables of statistics and peaks, and listening observations. Report units and achieved rates.
Comparison	Compare high- and low-rate results with the source reference and alias predictions on the same axes or in compact tables.
Discussion	Explain aliasing, DC/constant samples, leakage or transients where relevant, clipping/noise, repeatability limits, and playback limitations.
Conclusions	Answer the objective with numerical evidence and recommend suitable acquisition settings.
References and appendix	Cite course material. Place supporting plots, calculations, and code after the main report when needed.

Minimum analysis and evidence

- Give a compact acquisition table for every file: source, reference f , requested f_s , achieved f_s , N , duration, mean, RMS, strongest relevant FFT peak, and predicted alias when applicable.
- Use a common time span and consistent voltage scale when comparing waveforms. Show sample markers when low-rate sampling is the point of the figure.
- Use a common frequency range when comparing spectra. State whether the mean was removed and whether a window was used. Report $\Delta f = f_s/N = 1/T_{\text{record}}$.
- Include enough evidence to cover all three sounds and all five rates without placing fifteen nearly identical full-size plots in the main text. Use multi-panel figures or a summary table; move secondary plots to an appendix.
- For the tuning fork, compare the measured high-rate peak with the labeled frequency. Do not compare RMS in volts with frequency in hertz.
- Treat listening as qualitative evidence. Normalizing playback changes loudness, and resampling for the sound card cannot recover information lost during acquisition.

Equations and calculations to show

- Sample interval: $\Delta t = 1/f_s$
- Nyquist frequency: $f_N = f_s/2$
- Samples per cycle: f_s/f
- Record duration: $T_{\text{record}} = N/f_s$
- FFT-bin spacing: $\Delta f = f_s/N = 1/T_{\text{record}}$
- Mean: $\mu = (1/N) \sum x[n]$
- RMS: $x_{\text{RMS}} = \sqrt{(1/N) \sum x[n]^2}$
- Alias prediction: $f_{\text{alias}} = |f - k f_s|$, choosing integer k so $0 \leq f_{\text{alias}} \leq f_s/2$
- Relative difference: $100 | \text{measured} - \text{reference} | / | \text{reference} |$, used only for a meaningful nonzero reference

Figure and table requirements

- Number every figure and table and cite it in the text.

- Label both axes with quantity and unit. Use Microphone voltage (V), Time (s or ms), Frequency (Hz), and One-sided amplitude (V) where appropriate.
- Write a caption that states the sound source, sample rate, duration, and what the reader should notice.
- Use exported high-resolution figures, not screenshots of the MATLAB window.
- Make measured and predicted quantities visually distinguishable. Do not claim that connecting sampled points reveals the true path between samples.

Interpretation checks

Observation	What to check before concluding
A low-rate trace looks constant	Check its FFT including DC and use the phase/sample timing. A constant sequence is not proof of silence.
The RMS looks close to the reference	Inspect the spectrum. RMS measures overall voltage size and does not guarantee correct frequency content.
A peak spreads over bins	Consider finite record duration, off-bin frequency, and the chosen window before calling neighboring bins separate tones.
The replay pitch or duration is wrong	Verify that playback uses the achieved fs. Do not use N as the playback rate.
The waveform is flat or clipped	Check channel, cable, microphone excitation, gain/range, and source placement; repeat a known high-rate recording.

Final submission checklist

- The report answers all seven required questions and uses the group's own data.
- Every reported rate is identified as requested or achieved; every number has units.
- The text explains each important figure instead of asking the reader to interpret it alone.
- The abstract, results, and conclusion report consistent numerical findings.
- The PDF follows the current course template, opens correctly, and remains readable at normal size.
- All group members can explain the acquisition, calculations, MATLAB output, and conclusions.